

Vaibhav Dwivedi

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SUMMARY

Full-stack developer and AI/ML engineer with hands-on experience building GenAI applications, RAG pipelines, and scalable web platforms. Proficient in Python, JavaScript, LangChain, React, Node.js, Docker, and AWS. Delivers measurable impact through performance optimization, clean architecture, and production-grade engineering across distributed systems.

EXPERIENCE

ASSANJ

May 2026 – Present

Full Stack Developer

Remote

- Developing and optimizing **full-stack web applications** using **React, Node.js, and REST APIs**; collaborating in an agile, industry-oriented environment to ship features across the complete development lifecycle
- Implementing **scalable backend logic**, responsive UI components, and streamlined **CI/CD workflows** for real-world client-facing products; contributing to code reviews and architectural decisions

Analytx4t | [LOR](#)

Dec 2025 – Feb 2026

Generative AI Intern

Remote

- Built production GenAI applications using **LLMs, LangChain, and Python**; delivered **8 RAG pipelines** for document intelligence with vector search and tool-augmented agents
- Reduced inference latency by **40%** via chunking and embedding tuning; evaluated **5+ prompt strategies**, boosting RAG faithfulness scores by **22%** on internal benchmarks

PROJECTS

UniTalks (*Under Development*) | [Node.js](#), [Socket.IO](#), [React](#), [Redis](#), [WebRTC](#), [Docker](#), [AWS EC2](#), [Nginx](#) | [Video](#)

- Building a **full-stack anonymous real-time chat platform** at [unitalks.in](#); integrating WebRTC voice/video, Redis session state, and Nginx reverse proxy on AWS EC2 to support **1,000+ concurrent connections**
- Scaling horizontally via Redis Pub/Sub; containerized with Docker Compose and CI/CD via GitHub Actions, **reducing deployment time by 75%**; validated at **150+ concurrent users** under load testing

Baansinfra (*at ASSANJ*) | [React](#), [Node.js](#), [Supabase](#)

- Leading full-stack development of [baansinfra.com](#)—a project portfolio and lead-generation platform for a bamboo construction and architecture firm spanning residential, hospitality, and institutional builds
- Architected the backend with **Supabase** (PostgreSQL, Auth) for enquiry management and dynamic project categorization; built a responsive React frontend integrated with Node.js services

ResearchIQ | [Python](#), [LangChain](#), [QLoRA](#), [RAG](#), [FAISS/ChromaDB](#), [Semantic Search](#)

- Built an **AI-powered research intelligence platform** combining **fine-tuned LLMs**, Retrieval-Augmented Generation (RAG), and semantic search for automated paper analysis, abstract enhancement, and statistical recommendations
- Fine-tuned LLMs using **QLoRA** for domain-specific research tasks; engineered a RAG + semantic search pipeline enabling automated **research gap discovery** across large paper corpora

EDUCATION

Vellore Institute of Technology

2024 – 2028

B.Tech Computer Science – Specialization in Data Science | [CGPA: 8.10](#)

Chennai, India

- Impact:** Organized OpenHack 2025 (130+ participants, **\$5K+ prizes**); Event Coordinator at IE(I); Attended AI/ML workshops along with **200+ students** at IIT Madras and NIT Trichy; Programme Management Head at [Y-SoC](#), overseeing **500+ participants**
- Patent:** "System and Method for Stress and Pain Detection Using Multi-Scale Transformer-Based Neural Networks" – Application No. 202641065281 (Published)
- MSME:** Selected for the MSME 5.0 Program, VIT Chennai
- Hackathon:** Winning certificate, Hackwatts (Prodinno) – [Certificate](#)

Central Academy, Lucknow

Class 12 CBSE – PCM: 86% | *Class 10 CBSE – Overall: 92.3%*

SKILLS

Languages : C, C++, Java, Python, JavaScript (React.js, Node.js), TypeScript, HTML/CSS, Dart (Flutter), SQL

AI & ML : TensorFlow, PyTorch, Scikit-learn, LangChain, LangGraph, MLflow, Hugging Face, NumPy, Pandas, ChromaDB, FAISS

Web & Cloud : REST APIs, FastAPI, Next.js, Docker, AWS EC2, Redis, Supabase, PostgreSQL, Nginx, Vercel, Netlify, Render

Tools : Git, GitHub Actions, Jira, Figma, Power BI, Tableau, Postman, Linux

Certifications : MongoDB Building RAG Apps (2025) | Google Agentic AI Day (2025) | Microsoft Agentic AI Workshop (2025) | [AI Agents Workshop](#) (Kapidhwaj Innovations) – LLMs, LangChain & AI tooling